🔑 **Answer Key — Teacher Copy · Lesson 2-5**

MSTAR PRACTICE WORKSHEET

# Lesson 2-5: Describe Data by Range and Interquartile Range

Grade 6 Mathematics · Standard 6.DS.3

*These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.*

## Part A — Skills Check

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: B

A data set runs from a least value of 12 to a greatest value of 30. What is the range?

A. 12

**B. 18**

C. 21

D. 30

Range = greatest − least = 30 − 12 = 18.

**Question 2** SELECTED RESPONSE — CORRECT: A

On a box plot, Q1 = 20 and Q3 = 36. What is the interquartile range?

**A. 16**

B. 28

C. 36

D. 56

IQR = Q3 − Q1 = 36 − 20 = 16.

**Question 3** SELECTED RESPONSE — CORRECT: A

Which measure of spread uses ONLY the middle half of the data?

- (A) The interquartile range
- (B) The range
- (C) The median
- (D) The mode

> The IQR runs from Q1 to Q3, which is exactly the middle half of the values. The range is the distance from the minimum to the maximum; interior values do not enter that subtraction.

**Question 4** SELECTED RESPONSE — CORRECT: C

Two classes both have a median score of 81. Class A's IQR is 13 and Class B's IQR is 8. Which statement is true?

- (A) Class B has a higher median
- (B) Class A has more students
- (C) The middle half of Class B's scores is less spread out
- (D) Class B's highest score is greater

> A smaller IQR means the middle half of the data is packed into a shorter stretch, so Class B's middle half varies less. The medians are equal, so nothing there separates them.

**Question 5** SELECTED RESPONSE — CORRECT: C

A box plot shows: least 68, Q1 75.5, median 83.5, Q3 88, greatest 94. What are the range and the IQR?

- (A) Range 12.5, IQR 26
- (B) Range 20, IQR 12.5
- (C) Range 26, IQR 12.5
- (D) Range 26, IQR 83.5

> Range = 94 − 68 = 26. IQR = 88 − 75.5 = 12.5. The range covers the whole span; the IQR covers only the box.

**Question 6** SELECTED RESPONSE — CORRECT: B

Team A's salaries have a range of $14M and an IQR of $7M. Team B's have a range of $7M and an IQR of $2.5M. Both teams have the same median salary. What does this tell a player?

A Team B pays more overall

B Team B's salaries cluster much more tightly around the median

C Team A has more players

D Team B's highest salary is greater

Both measures of variation are smaller for Team B, so its salaries vary less: a typical player is more likely to earn close to the median. Team A spreads much wider in both directions.

## Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION

### Part A — correct answer: B

Two classes have the same median score of 81. Class A scores range from 78 to 84; Class B scores range from 60 to 99. What does this tell you?

(A) Class B has a higher median.

(B) Class B has greater variability, even though the centers are the same.

(C) Class A has more students.

(D) The two classes performed identically.

> Class A's range is 84 − 78 = 6; Class B's is 99 − 60 = 39. Equal centers can hide very different spreads, which is what measures of variability reveal.

- **A:** Both medians are 81, as the question states. What differs between the classes is how spread out the scores are.
- **C:** Neither range tells you how many students there are. A range describes spread, not count.
- **D:** The centers match, but Class A spans 6 points and Class B spans 39.

### Part B — correct answer: D

Why is the IQR often preferred over the range as a measure of spread?

(A) The IQR is always larger than the range.

(B) The range cannot be calculated for large data sets.

(C) The IQR uses every value in the data set.

(D) The range depends on only two values and is strongly affected by an outlier.

> The range uses only the minimum and the maximum, so one unusual value changes it completely. The IQR describes the middle 50% and is not moved by an extreme value.

**Question 8** WRITTEN RESPONSE · 2 POINTS

**Type II Reasoning: Same Center, Different Spread**

> Lena compared two data sets. Both had a median of 50. She concluded: 'The two sets are the same because their medians match.'

Explain why Lena's conclusion may be wrong, and name a measure that would tell the sets apart.

**Model answer:** The median describes the center only. One set could be 48, 49, 50, 51, 52 and another 10, 30, 50, 70, 90 — same median, very different spread. Range, IQR, or MAD would distinguish them.

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|---|---|
| **2 points** | Student explains that the median describes CENTER only, that two sets with the same center can have very different spreads, and names range, IQR, or MAD as a measure of variability. |
| **1 point** | Student says the sets may differ but does not name a measure of spread. |
| **0 points** | Student agrees with Lena or gives an unrelated response. |

## Part C — Show Your Thinking

Model answer and scoring guidance.

**Question 9** WRITTEN RESPONSE · 2 POINTS

Two students both scored 80 on a make-up test — one joins Class A (range 28, IQR 13) and one joins Class B (range 20, IQR 8). Both classes have a median of 81. Predict how each class's IQR changes, and explain your reasoning.

**Model answer:** Each new score of 80 lands just below the median of 81, which is inside the box for both classes. Adding a value inside the middle of the data pulls the quartiles slightly toward the center, so each IQR either stays the same or shrinks a little; neither can grow much, because nothing was added out at the extremes. The ranges do not change at all, since 80 is neither a new least nor a new greatest value for either class. The comparison holds: Class B's scores stay the more tightly clustered of the two.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.